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Soft subgrades' stabilization by using various fly ashes

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Abstract

This publication presents the results of research involving different types of self-cementing fly ashes (without any other activators) for the stabilization of four different types of soft subgrades from various road sites in Wisconsin, USA. The strength approaches were applied to estimate the optimum mixture design and to determine the thickness of the stabilized layer. The stabilized soil samples were prepared by mixing fly ash at different contents at varying water contents. The performance of fly ash stabilized subbase depends both on the specific source of fly ash and the engineering properties of soils. It is suggested that the performance analysis of fly ash should be based upon the laboratory tests such as index properties, compaction, unconfined compressive strength and CBR tests of a specific site. This is suggested rather than using the study of the physical properties and chemical composition of fly ash and soil. As disclosed in the literature, the strength gain due to stabilization depends mainly upon three factors; ash content, molding water content and compaction delay. The samples were subjected to unconfined compression strength and California bearing ratio (CBR) tests after 7 days curing time to develop water content–strength relationship. To evaluate the impact of compaction delay that commonly occurs in the field during construction, the sets of samples were compacted 2 h later after mixing with water. The unconfined compression strength and CBR tests were performed

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and used to determine the thickness of the stabilized layer in pavement design. All of these factors were taken into account throughout this research.

Keywords: CBR; Compaction; Fly ash stabilization; Soft subgrade; Unconfined compression strength

1. Introduction

Construction of roadways over soft subgrades is one of the most common problems for highway construction in many parts of the world as well as in the state of Wisconsin, USA. The usual approach to soft subgrade stabilization is to remove the soft soil, and replace it with a stronger material of crushed rock. The high cost of replacement has caused highway agencies to evaluate alternative methods of highway construction on soft subgrades. One approach is to use fly ash to stabilize the soft subgrade. This paper represents the results of laboratory investigation using four types of different fly ash for stabilization of four types of soft subgrades from road projects in Wisconsin, USA. The strength values of the soil–fly ash mixtures were evaluated to characterize the performance of stabilized soil as a road subbase. Unconfined compression strength and California bearing ratio (CBR) tests were performed to determine the strength properties of the soil–fly ash mixtures and the optimum mixture contents for construction. Stabilized soil specimens were prepared at 10–20% fly ash content (on dry weight basis) and different water contents. The mixture samples were compacted 2 h later after mixing with water, in order to evaluate the impact of compaction delay that commonly occurs in field construction. Therefore the 2 h compaction delay in the laboratory represents significant field conditions using the miniature Harvard compactor (36 mm in diameter and 71 mm long). The samples were subjected to unconfined compression strength tests. Additionally, CBR tests were performed on specimens compacted using the standard Proctor effort in a Proctor mold (152 mm in diameter and 178 mm long). Both the unconfined compression strength and CBR tests were performed after 7 days of curing in room temperature and 100% relative humidity in the humidity room. The effects of fly ash stabilization on strength properties are demonstrated in this research.

2. Fly ash

Fly ash is one of the most plentiful and versatile industrial by-products. It is generated in vast quantities (more than 69 million metric tonnes in 2002 in the USA) as a by-product of burning coal at electric power plants. Class C fly ash is usually recycled as an engineering material to take advantage of its pozzolanic characteristics. This type of fly ash provides the opportunity for applications where other activators would not be required. It offers more economical alternatives for a wide range of soil stabilization applications. The potential for using fly ash in soil stabilization has increased significantly in Wisconsin due to increased availability and the introduction of new environmental regulations (NR 538, Wisconsin Administrative Code) that encourage the use of fly ash in geotechnical applications when it is environmentally safe. Results of various investigations showed that soil stabilization

Table 1
The physical properties and chemical composition of fly ashes (Acosta, 2002)

Fly ash	Classification (ASTM C 618)	G_s	Percent fines	Moisture content (%)	LOI (%)	Lime (CaO) (%)	Other oxides ($SiO_2 + Al_2O_3 + FeO_3$) (%)
Class C[a]	–	–	–	3	6	24.3	50
Class F[a]	–	–	–	3	6	8.7	70
Columbia	C	2.70	95.3	0.09	0.7	23	55.5
Dewey	Off-spec	2.53	39.6	0.23	16.2	9.8	38.7
Edgewater	C	2.71	92.8	0.03	0.1	20.8	62.3
King	Off-spec	2.68	91.9	0.44	5.4	23.7	49.5

Notes: G_s, specific gravity; LOI, loss on ignition.
[a] After ASTM Standard C 618-01.

with Class C fly ash without any other activator is encouraging. The improved engineering properties of fly ash–stabilized soil are also reported (Edil et al., 2000, 2002; Senol et al., 2003; Turner, 1997).

Four types of fly ashes were used in this part of the main research: Columbia, Dewey, Edgewater and King. Columbia fly ash is from Unit-2 of the Columbia Power Plant in Portage, Wisconsin. Dewey fly ash is from the Nelson Dewey Power Plant in Cassville, Wisconsin. Edgewater fly ash is from Unit-5 of the Sheboygan Power Plant in Sheboygan, Wisconsin, and King fly ash is from the Allen S. King Plant in Bayport, Minnesota. Alliant Energy operates the Columbia, Dewey and Edgewater Plants. The King plant is operated by Xcel Energy. These fly ashes collectively provide a wide geographical coverage area as a source and also represent a wide range of characteristics for fly ashes available in Wisconsin. Columbia and Edgewater fly ashes are light brown in color, indicating higher calcium oxide content. On the other hand, Dewey and King fly ashes are dark gray and dark brown in color, indicating higher carbon content (Meyers et al., 1976).

Specific gravities and the percent fines for the fly ashes are essentially summarized in Table 1. The specific gravity of the Dewey fly ash is lower than the other three fly ashes. For the other fly ashes, the specific gravity ranged from 2.68 to 2.71. Columbia, Edgewater and King fly ashes have a powdery texture. Dewey fly ash has a more granular texture. The percent fines for Dewey fly ash is also lower than others at 39.6%. The other fly ashes have at least 92% fines. Columbia fly ash is slightly finer than the other ashes. The Columbia, Edgewater and King fly ashes have particles ranging from fine sand to silt and clay sizes, whereas Dewey fly ash has particles as large as medium sand and as small as clay (Meyers et al., 1976).

3. Laboratory tests

Various sets of the stabilized soil samples were prepared using 10, 12, 14, 16, 18 and 20% fly ash contents at different water contents for all types of soils and fly ashes. The mixture samples were compacted after 2 h with a compaction effort equivalent to standard Proctor, which were named 2-h delay samples. All the samples were allowed to cure for 7 days in room temperature and at 100% relative humidity conditions in the humidity room

(Acosta, 2002). The unconfined compressive strength and CBR values of the soil samples were determined and were found to substantially increase as a result of stabilization. The effects of fly ash content, water content and 2 h compaction delays were also studied and it was found that all three factors have a significant impact on the performance of stabilized soil. ASTM standards were followed on all of the tests performed for this study.

3.1. Engineering properties of subgrades

In Wisconsin approximately 60% of the surficial soils provide poor subgrade stability. Four weak subgrades were used in the testing program for this research. The sampling locations of soils were identified by the Wisconsin Department of Transportation and intended to represent the range of soft subgrades typically encountered in Wisconsin. The samples at each site were collected along the highway shoulder, at a depth of 0.6–0.9 m (Edil et al., 2000, 2002).

The index and compaction properties and classifications of the soils are summarized in Table 2. The Atterberg limit tests were performed and the liquid and plastic limits were determined. All of the soils were fine-grained materials and classified as clays or silt according to the USCS and AASHTO. Based on the test results the USCS classification of the Scenic Edge soil was low plasticity clay (CL), the STH60 Lodi soil was low plasticity silt (ML), the STH28 Mayville soil was high plasticity organic soil (OH), and the USH151 Platteville soil was high plasticity clay (CH). The AASHTO classification of soils of all sites was found as clay. These essentially inorganic clays had loss on ignition (LOI) values less than 10%. All of the soils contained at least 96% fines. The test results as well as the classification are tabulated in Table 2.

Also, the compaction tests were performed to get the optimum water content and maximum dry unit weight of soil samples of each site. Typical bell-shaped compaction curves were obtained and the results of samples of all sites are illustrated in Fig. 1. To determine the compaction properties of the mixture samples, a Harvard Miniature Compactor was used similar to the standard Proctor test. The compaction curves for stabilized soils were determined in terms of using the Harvard Miniature Compaction procedure (ASTM D4609-01).

3.2. Engineering properties of stabilized soils

3.2.1. Compaction tests

For the subbase condition, the samples were prepared approximately 7% wetter than the optimum water content. These specimens were prepared to simulate the natural wet condition observed in the field during the rainy season. The compaction curve corresponding to the standard Proctor effort was determined for each soil specimen following the procedure in ASTM D 698-00a. Air-dried soil that passed a US no. 20 standard sieve was mixed homogeneously with the required percent of fly ash. Then the required amount of water was sprayed on the soil–fly ash mixture. All mixtures were prepared with fly ash contents between 10 and 20% on dry weight basis with the soil. The sets of mixtures were compacted 2 h after mixing with water (2-h delay) to simulate the typical duration in a mold (35 mm in diameter and 70 mm height) that occurs in the field. After adding fly ash to the subgrade,

Table 2
Index properties, compaction characteristics, classification and CBR of soils

Sampling location	LL (%)	PI (%)	Fines (%)	LOI (%)	Classification		CBR	w (%)	γ_d (kN/m^3)	w_{OPT} (%)
					USCS	AASHTO				
Scenic Edge, Cross Plains, WI	44	20	96	2	CL	A-7-6	1	27	16.2	20
STH 60 Lodi, WI	39	15	96	1	ML	A-6	3	25	16.5	19
STH 28 Mayville, WI	61	19	97	10	OH	A-7-5	0.3	35	13.5	29
USH 151 Platteville, WI	60	35	97	4	CH	A-7-6	0.4	32	16.4	19

Notes: percent fines, percentage passing no. 200 sieve; LOI, loss on ignition; CBR, California bearing ratio (performed approximately 7% wet of optimum water content).

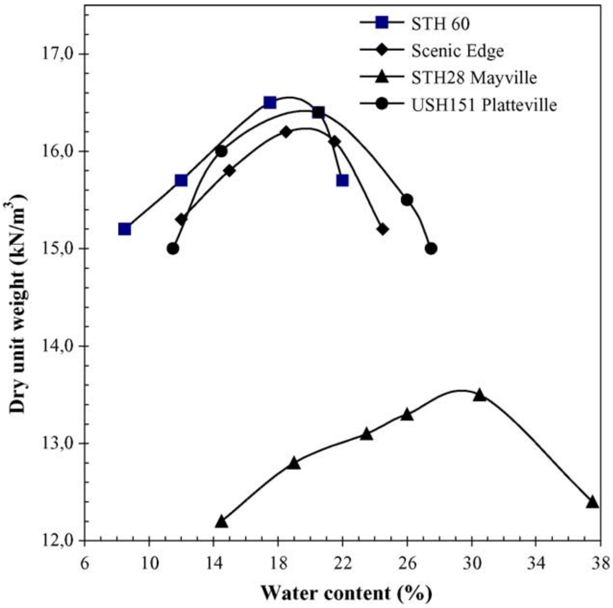

Fig. 1. The compaction curves of the soils (Acosta, 2002).

the engineering properties definitely started to change and the maximum dry unit weight for "2-h delay" stabilized soil was lower than that for the soil alone. The optimum water content was slightly higher (1%). The relation between the dry unit weight of all mixture samples and fly ash contents is shown in Fig. 2. Consequently, the maximum dry unit weight decreased and the optimum water content increased when the fly ash content increased. Only the USH151 soils with king fly ash mixture samples did not show that behavior for the fly ash content between 10 and 18%, based on the test results.

3.2.2. Unconfined compressive strength and CBR tests

Based on geotechnical engineering literature, the stabilization of soil by using Class C fly ash significantly changes and improves the engineering properties of soils such as unconfined compressive strength, compressibility, durability and permeability. A compaction delay of 2 h was used in all the laboratory tests, in order to determine the effects of compaction delay to simulate the field conditions. Two different types of strength tests were performed to determine the strength parameters. These were unconfined compressive (q_u) test and CBR test.

The unconfined compressive strength of soil is considerably one of the most important designing parameters used for pavement design for city streets and highway construction. The unconfined compression tests were conducted both on in situ soil and stabilized soil and the results are given in Table 3. Before the unconfined compression test of the stabilized soil, the soils were air-dried and all the clods were crushed properly. The soil passing through the #20 standard sieves was collected. The samples were then compacted using the Harvard

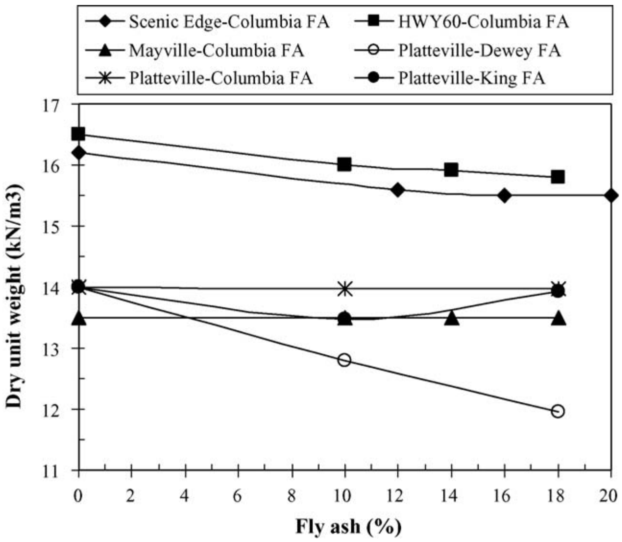

Fig. 2. The comparison of fly ash content with dry unit weight values.

Compactor based on ASTM D698-00a. The 2-h delay samples were compacted after 2 h from the time of mixing with water. After compaction, each sample was placed immediately in a polyethylene bag to prevent any desiccation. They were then cured for 7 days according to ASTM Standard Procedure D1632-96 (that is, protected from the free moisture in 100% relative humidity room at 21 °C).

Table 3
Unconfined compressive strength (kPa) of soil and soil–fly ash mixtures

Soil name	FA (%)	Scenic Edge, WI	STH 60 Lodi, WI	STH 28 Mayville, WI	USH 151 Platteville, WI
Optimum water content of soil (w_{OPT})					
Soil	0	140 (−1.0)	133 (2.0)	190 (−0.9)	446 (−1.2)
Columbia	10	–	566 (1.9)	360	–
	12	772 (2.4)	–	–	–
	14	–	614 (2.7)	465 (−1.0)	–
	16	828 (3.2)	–	–	–
	18	–	649 (3.4)	516 (−1.0)	–
	20	863 (4.0)	–	–	–
7% Wet of w_{OPT}					
Soil	0	62 (7.0)	90 (7.0)	32 (5.4)	106 (7.3)
Columbia	10	–	–	–	356 (7.0)
	18	–	–	–	450 (7.0)
Dewey	10	–	–	–	312 (7.0)
	18	–	–	–	477 (7.0)
King	10	–	–	–	244 (7.0)
	18	–	–	–	348 (7.0)

Note: specimens of soil alone tested immediately after compaction.

The cured samples were tested for unconfined compressive strength according to ASTM D2166-00. The tests were performed to develop a water content–strength relationship and to determine the effect of compaction. It was determined that the fly ash stabilization increased the strength significantly for four types of soils. The strength behavior of the mixtures depended on the engineering properties of the soils. In addition, the physical properties of fly ash are very important factors for the mixtures. It was observed that the maximum strength increased with increasing the fly ash content for the types of soils studied in this research, but it may not be the same for all types of soils.

For CBR tests, the air-dried soil that passed a US no.4 standard sieve was mixed homogeneously with the pre-determined amount of fly ash, and then the pre-determined amount of water was sprayed on soil–fly ash mixture. Similar to the unconfined compressive strength tests, the soil–fly ash mixtures with water were kept in an airtight polythene bag and were compacted in a standard CBR mold 2 h after mixing with water. The CBR tests were performed on stabilized soils at different fly ash contents and at a molding water content of 1% wet of optimum, which corresponded to the maximum unconfined compressive strength. Also, some specimens were prepared at near the optimum and also 7% wet side of the optimum water content by using the standard Proctor compaction effort to simulate the optimum and wet conditions typically observed. The soil–fly ash mixture samples were left in the mold, sealed using a plastic wrap, and left to cure for 7 days at 21 °C and 100% relative humidity prior before they were tested. Then the CBR tests were performed in accordance with ASTM D1883-99. Similar to the unconfined compressive strength values, the CBR values increased with increasing fly ash content for some types of soils and the rate of increase of CBR values was found to diminish as the fly ash content increased (Senol et al., 2003).

4. The evaluation of results of laboratory mixture samples of unconfined compressive strength and CBR tests

A general trend of increasing unconfined compressive strength with increasing fly ash content was observed as shown in Fig. 3. Also, the unconfined compressive strength data are given in Table 3. In Fig. 3, the graphic of six mixture variations of four types of subgrades with three types of fly ashes illustrates that the strength gain was achieved in terms of soil type, fly ash type and water content. However, it was not strongly correlated with soil index properties. Essentially it was observed that soft subgrade mixtures with fly ash improve the strength properties of soils. Depending on the physical properties of Columbia fly ash, it provides results better than other types of fly ashes for all soil mixtures prepared at optimum water contents used in this research. The mixtures of low plasticity clay (CL) Scenic Edge and Columbia fly ash had the highest unconfined compression strength values. The mixtures of low plasticity silt (ML) STH60 and Columbia fly ash had also high strength values. This low plasticity, fine materials had higher unconfined compressive strength values than the other two high plasticity, fine materials. The unconfined compressive strength values for the Mayville soil (OH) and Platteville soil (CH) mixtures with Columbia fly ash were similar. The other investigation was mixtures of Platteville soils (CH) with three different types of fly ashes. The unconfined compressive strength values of mixtures with Columbia fly

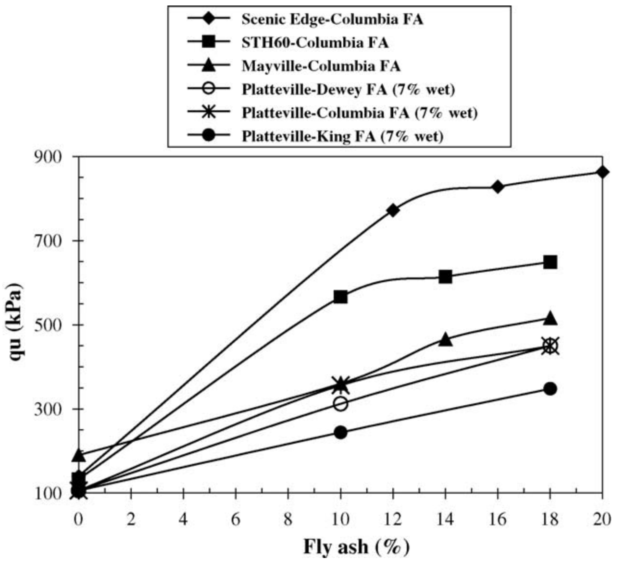

Fig. 3. The comparison of fly ash content with q_u values.

Table 4
CBR of soil and soil–fly ash mixtures compacted 2-h after mixing and cured for 7 days prior to the CBR test

Soil name	FA (%)	Scenic Edge, WI	STH 60 Lodi, WI	STH 28 Mayville, WI	USH 151 Platteville, WI
Optimum water content of soil (w_{OPT})					
Soil	0	3 (4.0)	5 (1.0)	2 (0.5)	17 (1.0)
Columbia	10	–	32 (2.0)	2 (2.0)	–
	12	34 (2.4)	–	–	–
	14	–	36 (2.7)	–	–
	16	51 (3.2)	–	–	–
	18	–	38 (3.4)	5 (4.0)	–
	20	56 (4.0)	–	–	–
Dewey	10	–	–	4 (2.0)	20 (2.0)
	14	–	–	–	20 (3.0)
	18	–	–	10 (4.0)	25 (3.0)
Edgewater	10	–	–	2 (2.0)	–
	18	–	–	2 (4.0)	–
7% Wet of w_{OPT}					
Soil	0	1 (7.0)	3 (6.0)	0.3 (6.9)	3 (5.0)
Columbia	10	–	–	–	12 (7.0)
	18	–	–	–	15 (7.0)
Dewey	10	–	–	–	10 (7.0)
	18	–	–	–	31 (7.0)
King	10	–	–	–	9 (7.0)
	18	–	–	–	20 (7.0)

Notes: STH28 Mayville ($w_{OPT} = 29\%$), USH 151 Platteville ($w_{OPT} = 19\%$), STH 60 Lodi ($w_{OPT} = 19\%$), Scenic Edge ($w_{OPT} = 20\%$). Number in parenthesis indicates the water content of the soil relative to the soil optimum water content ($w_{SOIL} - w_{OPT}$).

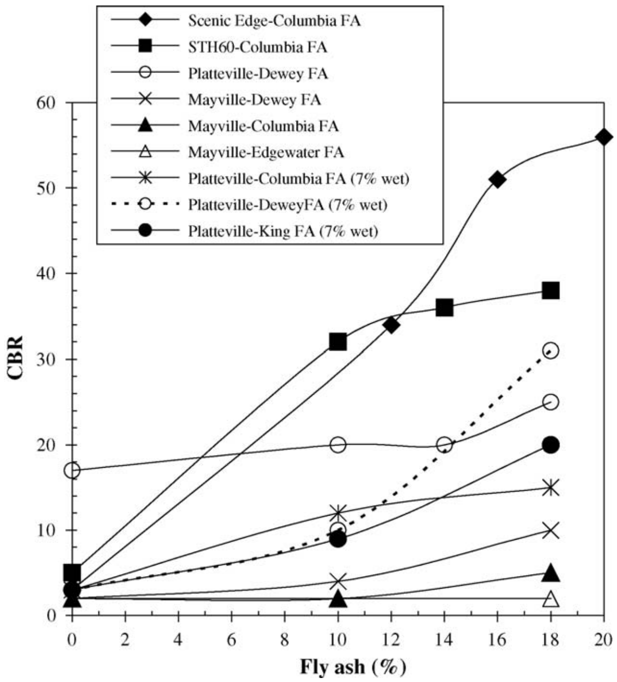

Fig. 4. The comparison of fly ash content with CBR values.

ash were comparable to the mixture values with Dewey fly ash. The lowest unconfined compressive strength values were from Platteville soil with King fly ash. All Platteville soil mixtures with various fly ashes had the lowest strength values because the water content of all mixtures were 7% wetter than the optimum water content. It proves that higher water content causes a decrease in the strength of the subbase.

CBR values are widely used to design the base and subbase layer for the pavement construction. Air-dried samples were sieved through #10 standard sieve before they were used. To determine the CBR of the native soil, one subgrade sample without fly ash was tested in its natural condition, close to natural water content. The CBR results of the soils and mixtures with fly ash are given in Table 4.

A general trend of increasing CBR values with increasing fly ash content was observed. The gain in CBR values depended on the amount of fly ash and water content in the mixture. The soil–fly ash mixtures of all sites were prepared for 10 and 20% fly ash contents and compacted at the optimum water content and also at 7% wetter than optimum.

High plasticity fine material subgrades such as the Scenic Edge and STH60 mixture samples provided the higher CBR results than the other samples. Columbia fly ash was used with these samples. High plasticity Platteville clay samples were mixed with various types of fly ashes at close to 7% wetter than the optimum water content and with the Dewey fly ash at about the optimum water content. Furthermore, soils that were compacted close to their optimum water contents generally had a higher CBR value than when they were compacted 7% wetter than the optimum water content as shown in Fig. 4. Mixtures with

Dewey fly ash also showed that the mixtures compacted with the optimum water content generally had higher CBR value than the same mixture compacted 7% wetter than the optimum water content. This indicates the effect of water content. Only the 18% Dewey fly ash mixtures 7% wetter than the optimum water content, had a higher CBR values than the mixture compacted at the optimum water content. Finally, the high plasticity organic Mayville soils had the lowest CBR values for various fly ash mixtures at the optimum water content. The organic soil or high plasticity clay samples studied in this research had lower CBR value results for fly ash mixtures than the results of low plasticity clay or silt samples. The presence of organic matter (10%) in one of the soils (STH 28 Mayville) may have inhibited the pozzolanic reaction between fly ash and water, resulting in smaller increase in CBR values compared to the inorganic soils.

5. Results and conclusions

The objective of this research was to quantify the effect of fly ash stabilization on four different types of soft subgrades encountered using locally available fly ash in Wisconsin. The improvement in engineering properties of soft subgrades such as unconfined compressive strength and CBR was investigated. Soil–fly ash mixtures were prepared at different fly ash contents (10–20%) with the specimens compacted at the optimum water content and about 7% wetter than the optimum water content. Unconfined compression strength and CBR tests were then performed on these mixtures. The fly ash stabilization increased both the unconfined compressive strength, and the CBR values substantially for the mixtures tested and have the potential to offer an alternative for soft subgrade improvement of highway construction. The research has shown that stabilizing the soft subgrade at specified water contents and minimizing compaction delay in the field could maximize the strength of fly ash-stabilized soils.

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